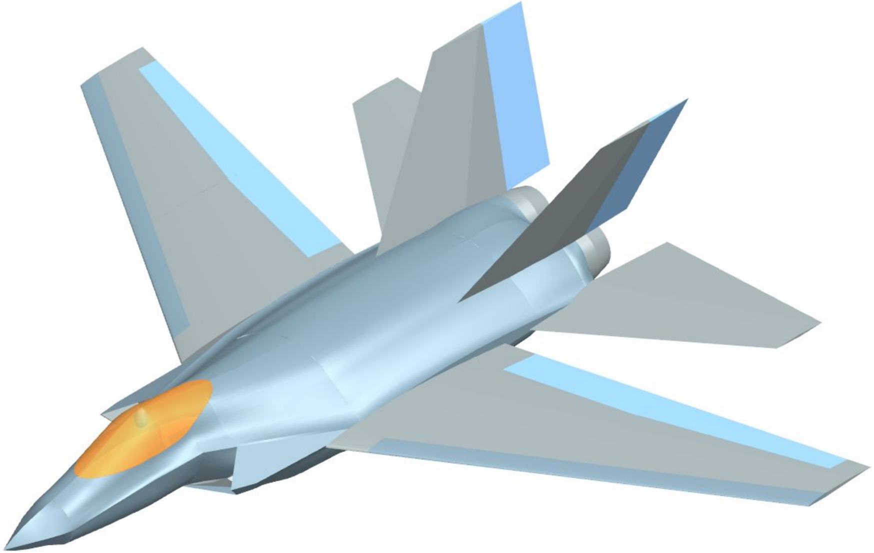
F/A-67 Preliminary Design Review



Group 01- No Thai Grubbin

Michael Chen, Charles Choi, Cristina Erskine, James Gold,
Santiago Ramos-Assam, Thomas Sheridan

October 17, 2025

Request for Proposal

Next Generation Carrier-Based Strike Fighter

- Replacement for F/A-18E/F aircraft
- 500 units with 2035 initial operating capability
- 25-year service life

General Design Requirements

- Use existing production engines and currently available materials
- Technology readiness level (TRL) 6 subsystems
- Internal carriage of ordnance desired

Objective

- Maximize combat performance with comparable cost to F/A-18E/F
- Flyaway cost, 500 unit production run: <56.5 $B 2025

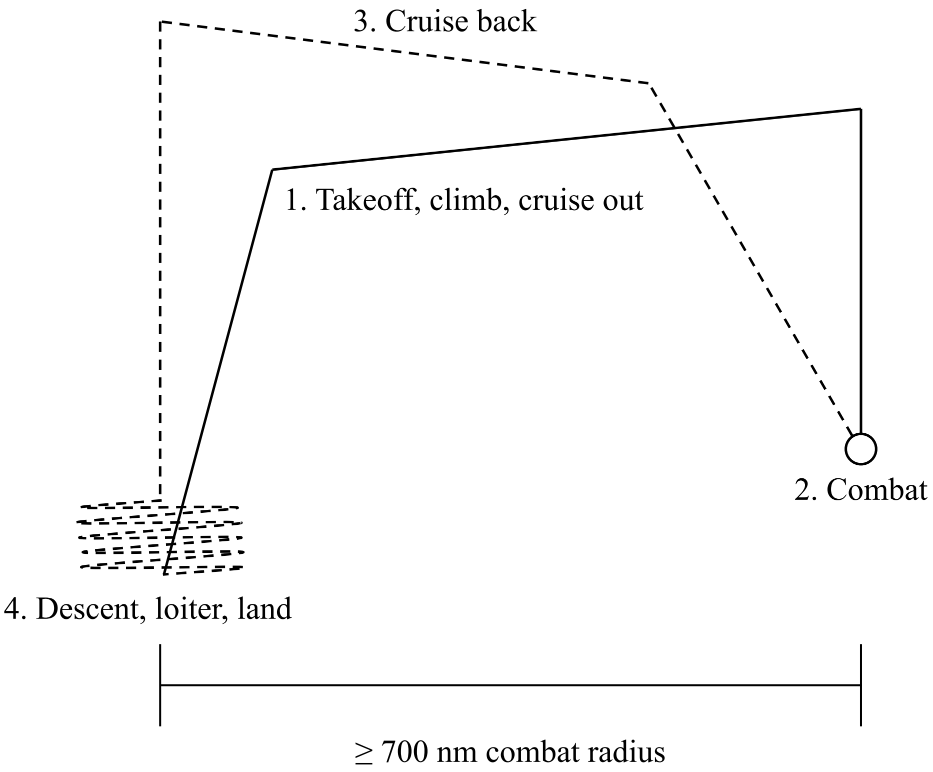
Air to Air Combat Mission

Minimum Performance Requirements

- 700 nm combat radius
- 2 min combat at maximum thrust (10,000 ft)
- 8 deg/s sustained turn rate (20,000 ft)
- Mach 1.6 dash speed (30,000 ft)
- 7g vertical load factor (mid mission weight)

Payload

- 6x AIM-120C
- 2x AIM-9X



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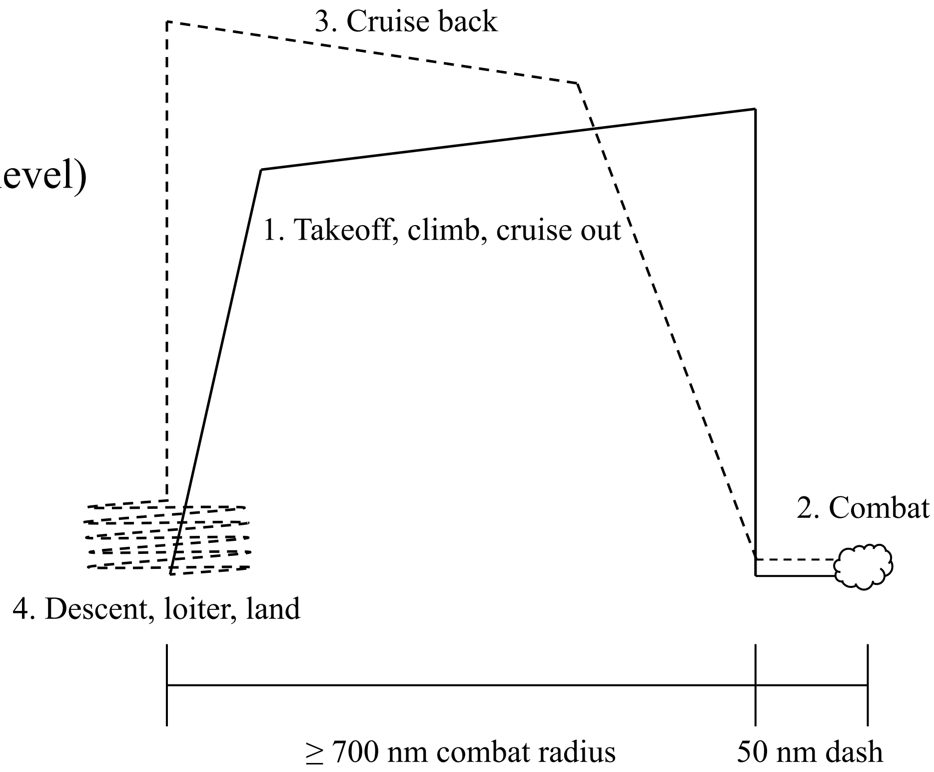
Strike Mission

Mission Performance Requirements

- 700 nm combat radius
- 2x 50 nm dashes at intermediate thrust (sea level)
- Mach 0.85 dash speed (sea level)
- 7g vertical load factor (mid mission weight)

Payload

- 4x MK-83 JDAM
- 2x AIM-9X



Design Summary

Dimensions

- Length: 50 ft
- Span: 60 ft, Stowed span: 35 ft
- Height: 18.5 ft
- Planform area: 670 ft^2
- Spot factor: 1.17

Weights

- Empty weight: 41,432 lb
- Payload: 2,460 - 4,424 lb
- Fuel: 32,274 lb
- Takeoff weight: 76,366 lb
- Avionics: 2500 lb

Configuration

Crew: 1

Wing:

- Swept
- Supercritical airfoil
- Leading edge flaps
- Slotted flaps and flaperons

Empennage:

- Swept, biconvex airfoil
- All moving horizontal stabilator
- Twin vertical stabilizers

Power Plant:

- 2x F110-GE-132

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Mission Performance and Weights

Air to air

- 900 nm combat radius
- 2.5 min combat time
- <u>11 deg/s</u> sustained turn rate
- Mach 1.7 dash speed
- <u>8g</u> vertical load factor

Strike

- <u>1020 nm</u> combat radius
- <u>Mach 0.9</u> dash speed
- <u>8g</u> vertical load factor

	Air to air	Strike
Empty weight (lbs)	41,432	41,397
Payload weight (lbs)	2,460	4,424
Fuel weight (lbs)	32,274	30,270
Takeoff weight (lbs)	76,366	76,292

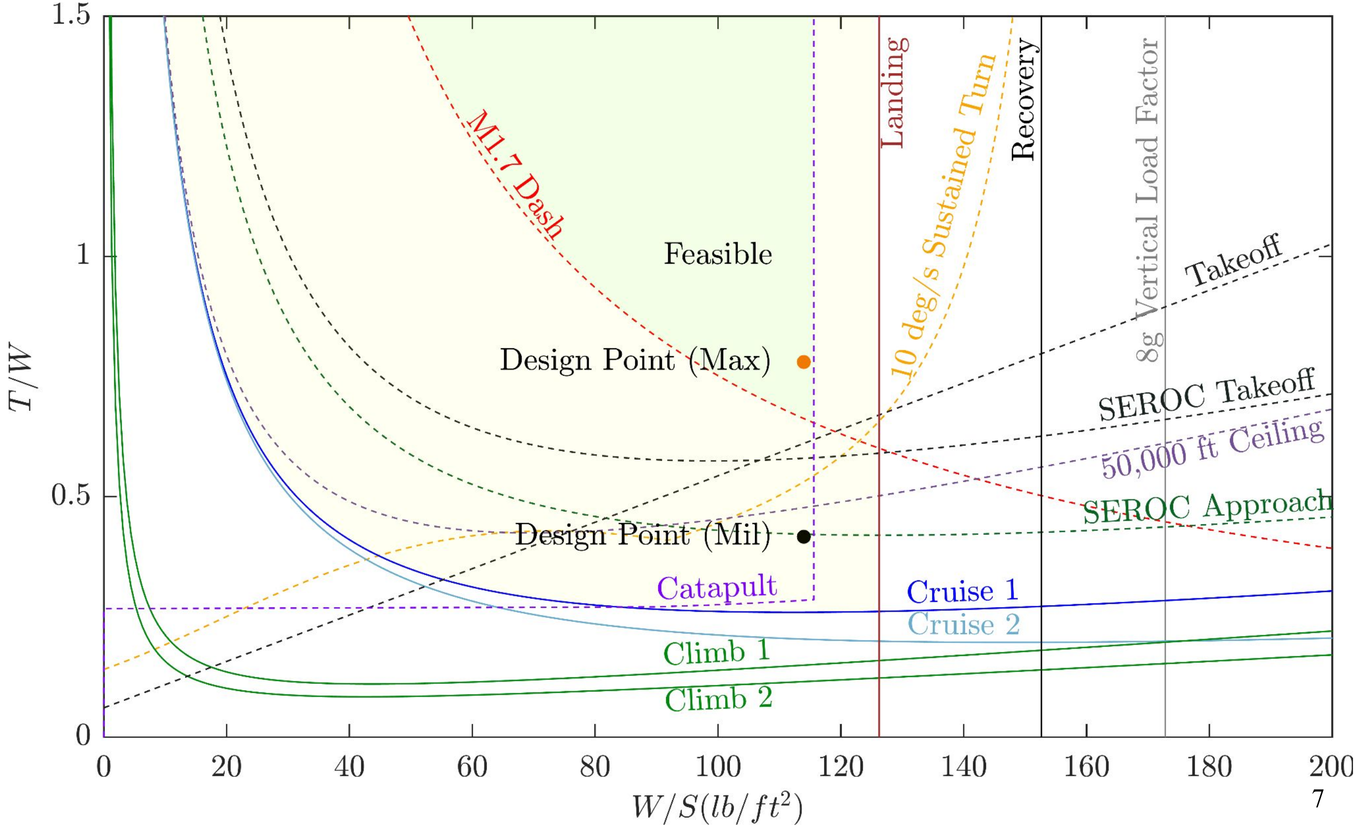
Underlined parameters meet desired RFP values
All parameters exceed minimum requirements

Thrust-to-Weight vs. Wing Loading Sizing Plot (Air to Air)

W/S = 114 lb/ft^2
T/W (max) = 0.78
T/W (mil) = 0.42

Constrained by dash and catapult launch

Design point meets afterburning and dry T/W requirement



Thrust-to-Weight vs. Wing Loading Sizing Plot (Strike)

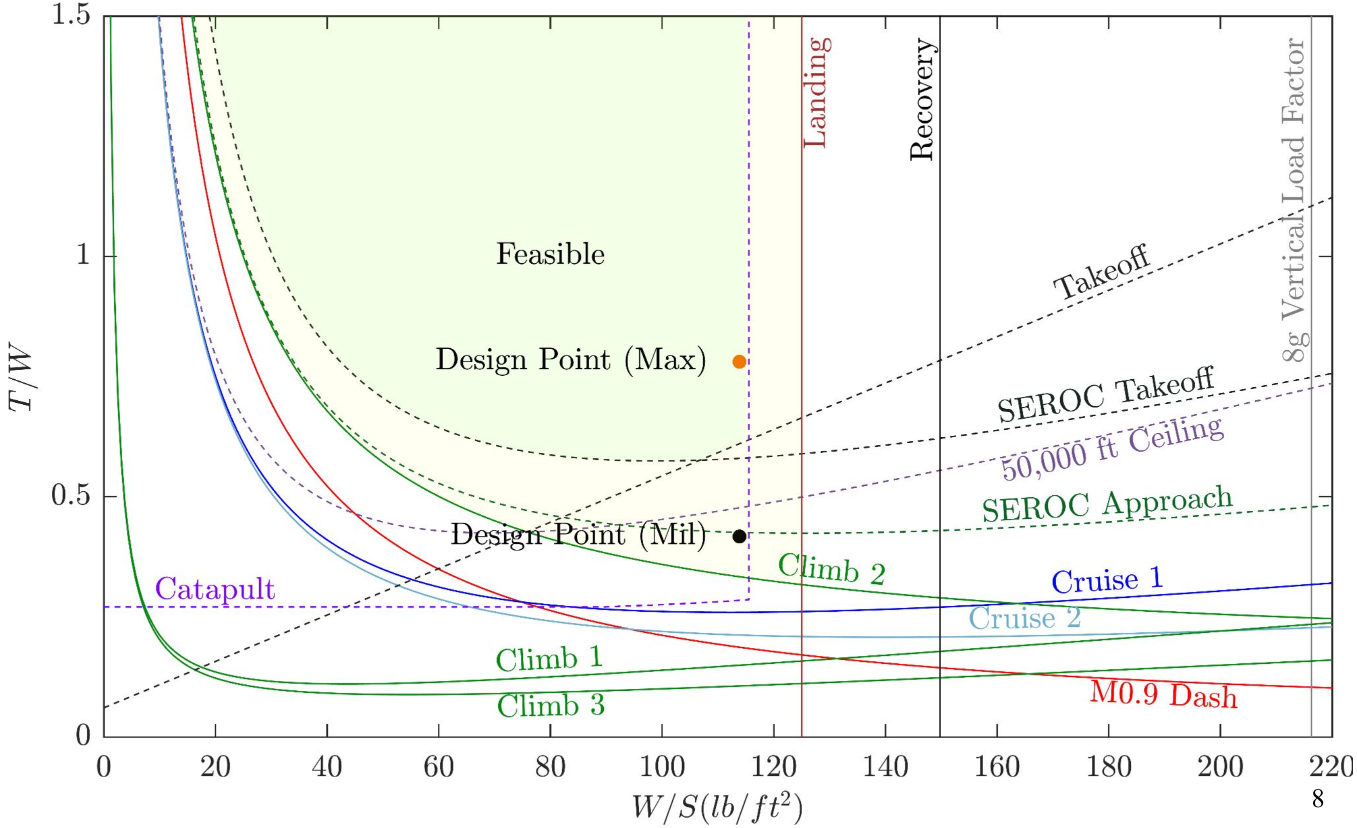
W/S = 114 lb/ft^2

T/W (max) = 0.78

T/W (mil) = 0.42

Constrained by dash, catapult launch, and conventional takeoff

Design point meets afterburning and dry T/W requirement

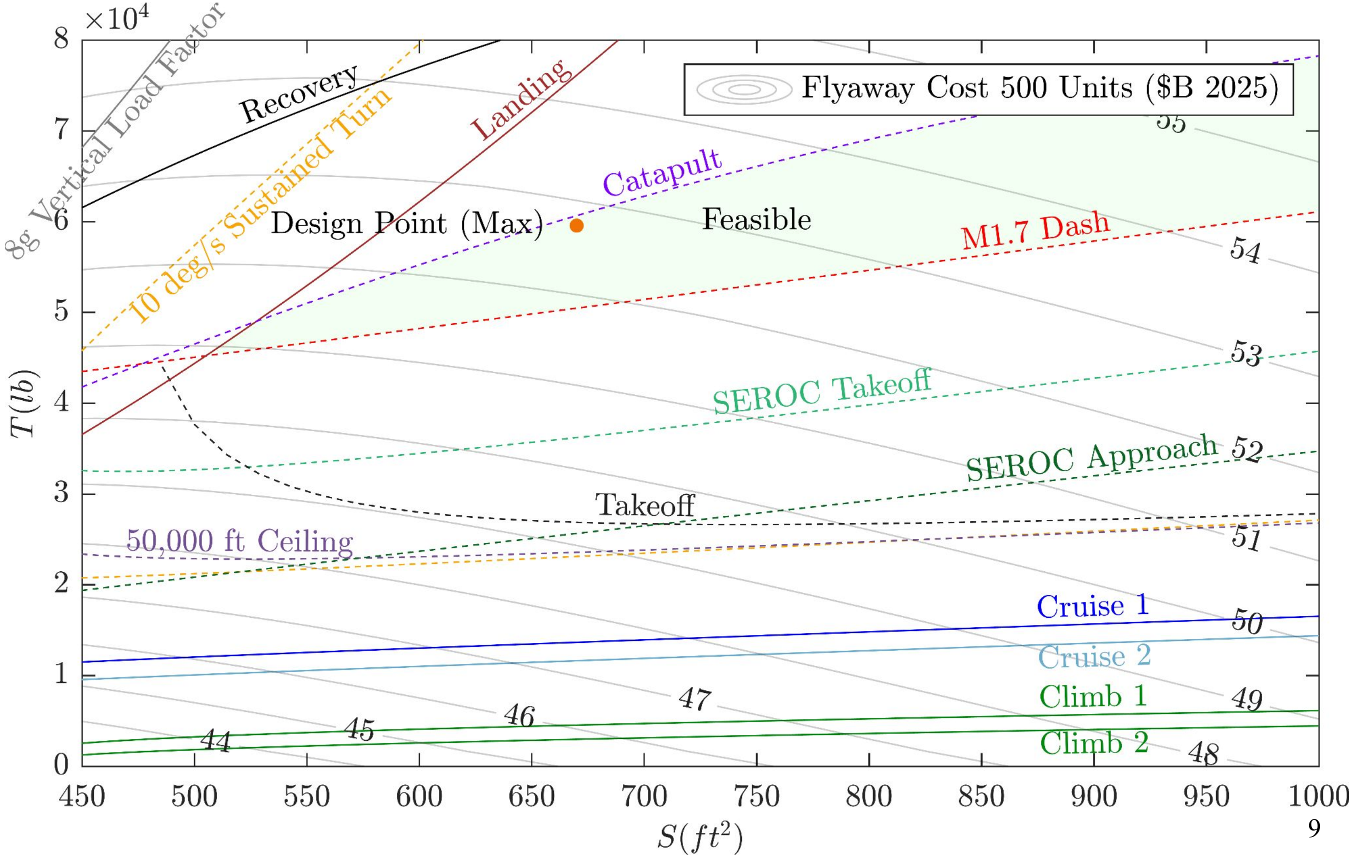


Thrust vs Wing Area Sizing Plot (Air to Air)

S = 670 ft^2
T = 29,784 lb

Assumes 10% thrust loss from intake and engine offtakes

Design point meets afterburning thrust and cost requirement



Thrust vs Wing Area Sizing Plot (Strike)

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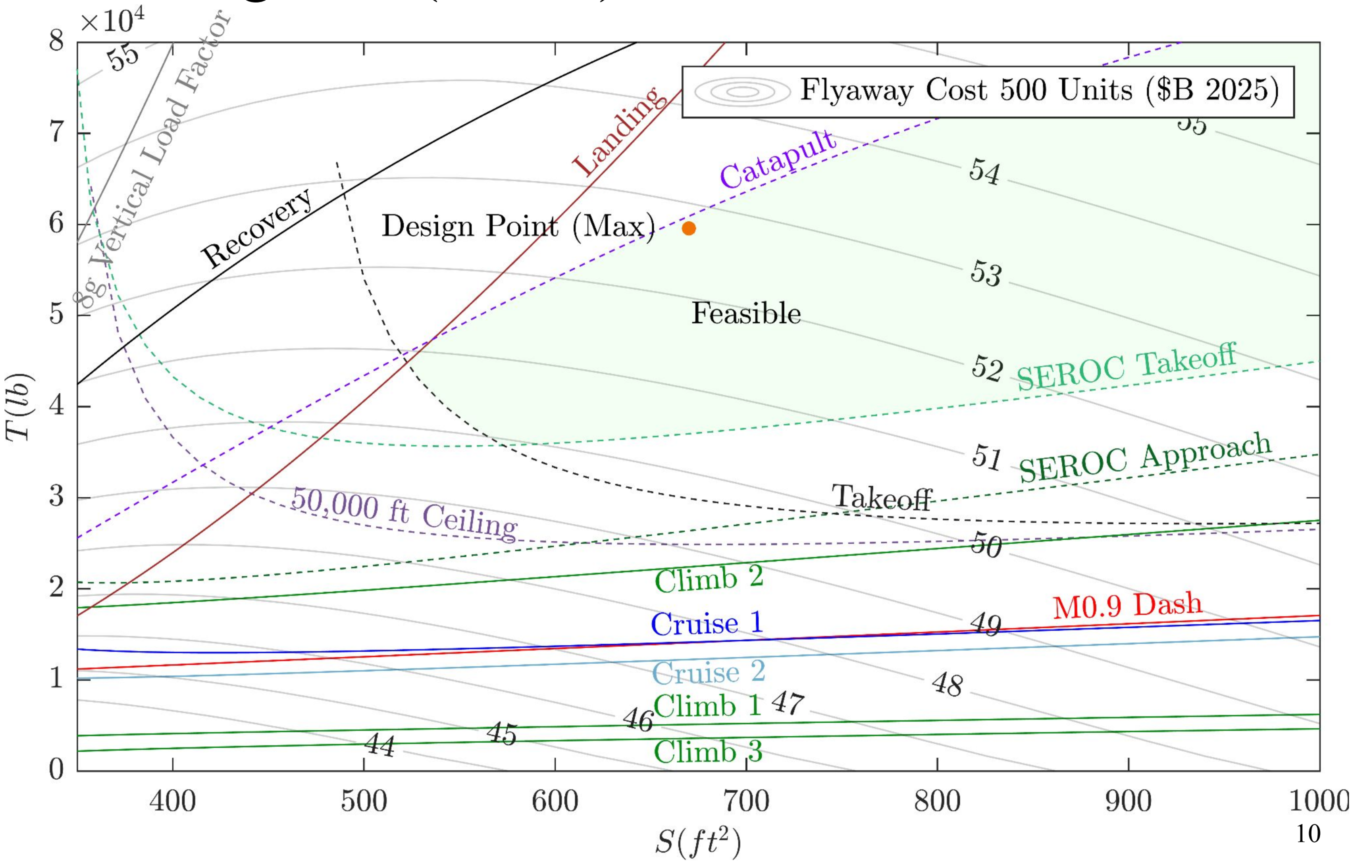
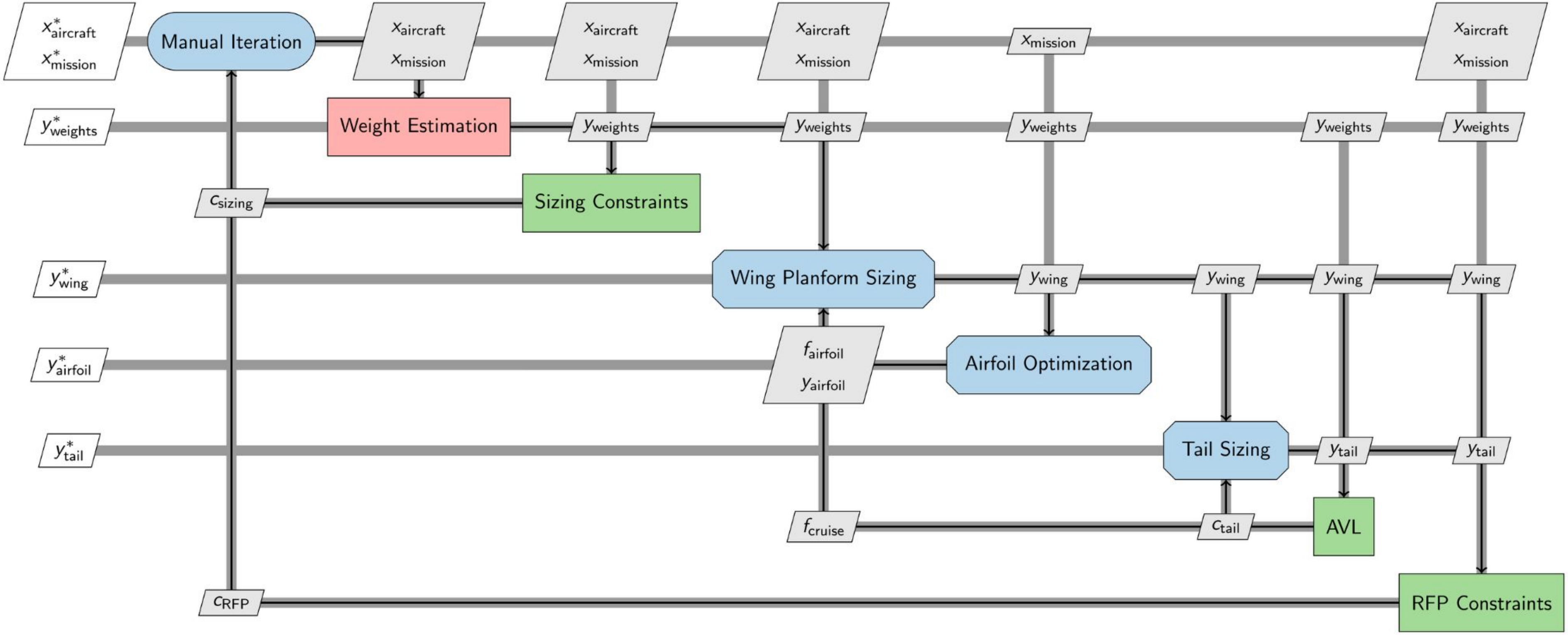


Computation Procedure



[1]

Configuration Selection

Preliminary Sizing Trends

- Higher mission performance increases takeoff weight
- Higher aspect ratio (AR) and lift-to-drag ratio (L/D) decrease takeoff weight

RFP Design Drivers

- Catapult launch requirements limit takeoff weight
- Aircraft dimensions requirements limit span and planform area
- Dash and Takeoff constraints require higher afterburning T/W

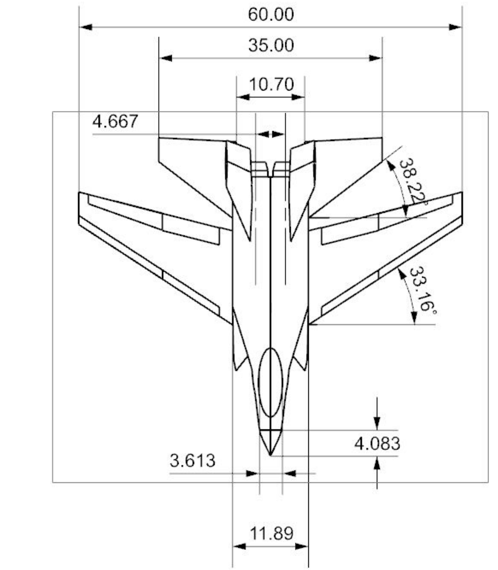
→

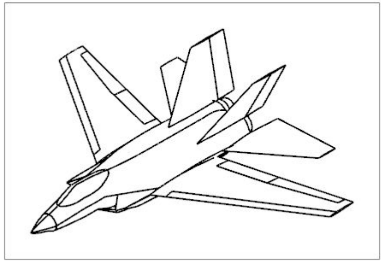
Design for Objective

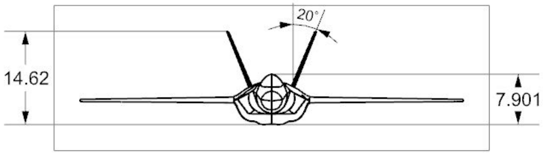
- Maximizing span increases mission performance
- Slotted flaps increase $C_{L,max}$, increasing catapult launch takeoff weight
- Wing area minimized to reduce aircraft size and weight
- 2 engines selected to provide sufficient T/W

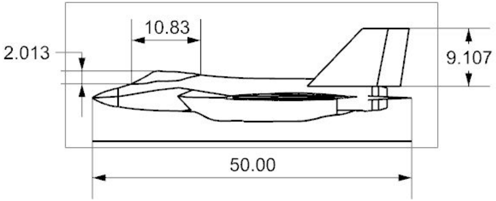
Configuration selected to maximize mission performance

Drawing and Dimensions



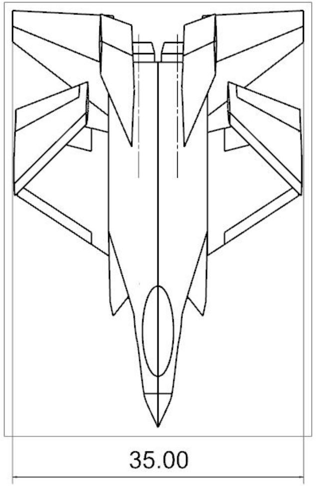


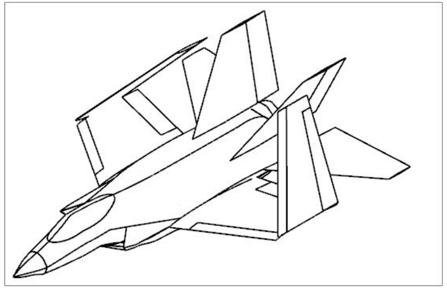


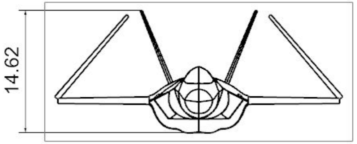


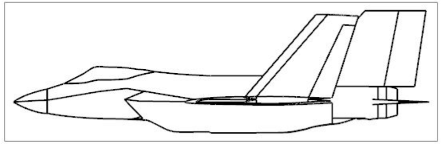
Scale 1:360	Dimensions are in feet

Drawing and Dimensions: Stowed Configuration









Scale 1:280	Dimensions are in feet

Comparison With Existing Aircraft

Aircraft	Empty Weight (lb)	Takeoff Weight (lb)	Length (ft)	Span (ft)	Reference Area (ft^2)	Aspect Ratio	T/W	W/S (lb/ft^2)	Combat Radius (nm)	Flyaway Cost ($B 2025)
F/A-67	**41,432**	**76,366**	**50.0**	**60.0**	**670**	**5.4**	**0.78**	**114**	**1020**	**52.5**
F/A-18E	32,081	66,000	60.2	44.7	500	3.5	0.65	132	462	56.5
F-35C	34,800	70,000	51.5	43.0	668	2.7	0.61	105	670	63.5
F-15E	37,500	81,000	63.8	42.8	608	3.0	0.73	133	687	30.7

Conclusions

- Higher wingspan selected to increase cruise performance and combat radius
- Reference area constrained by RFP dimension requirements
- F/A-67 flyaway cost less than F/A-18E

Notes

- Comparable takeoff weights are MTOWs
- Combat radius is maximum for similar mission profiles
- Flyaway cost is for 500 units

Wing Planform Design

Geometry

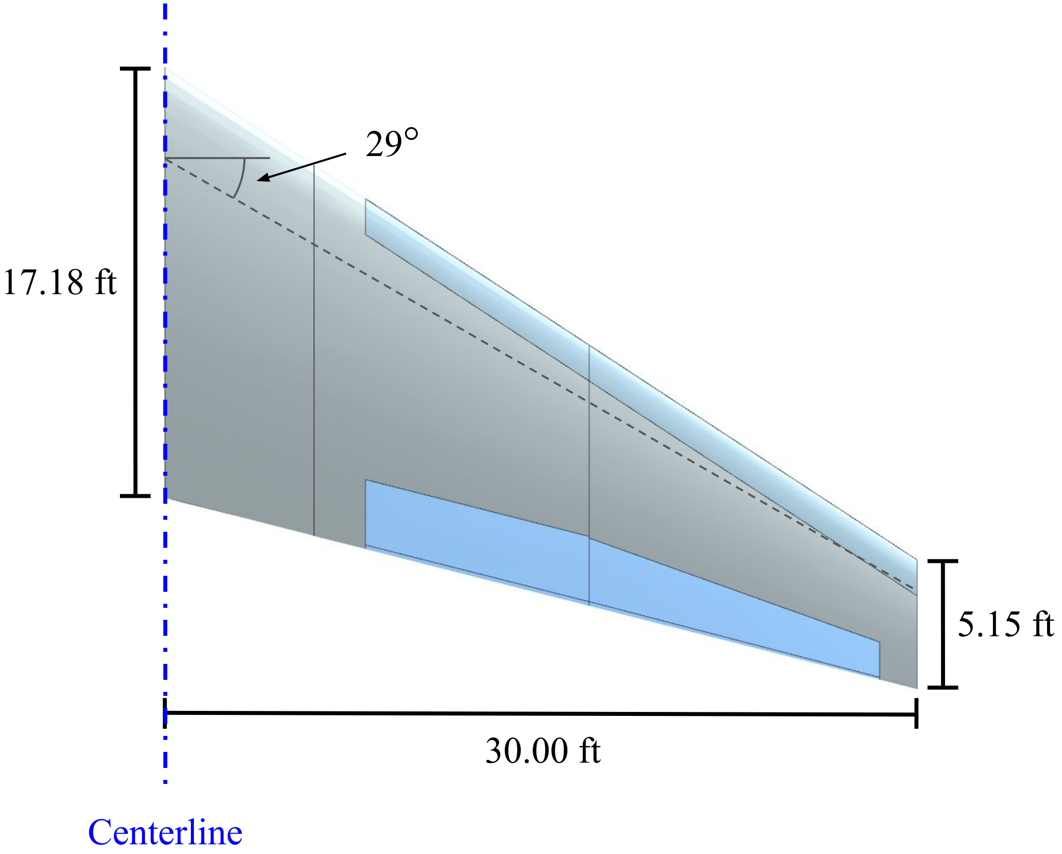
- Tapered to reduced induced drag
- Swept to reduce transonic wave drag
- Twisted to achieve high efficiency

Dimensions

- Span: 60 ft
- Reference Area: 670 ft^2
- MAC: 12.24 ft
- Aspect Ratio: 5.37
- Taper Ratio: 0.3
- Sweep: 29 deg
- Twist: -2.7 deg

Cruise Performance

- $C_{D,ind} = 0.01843$, $C_{D,visc} = 0.01205$, $C_{D,wave} = 4.04 \times 10^{-5}$



Wing Planform Trade Study

Procedure

- Minimize total cruise drag
- Vary taper ratio, sweep angle, twist angle

Assumptions

- Zero root incidence
- Carry-through wing (no fuselage or tail)
- Korn wave drag (supercritical airfoil)

Results

- Taper ratio of 0.3 minimizes parasitic drag
- 29 degree sweep optimal, higher sweep increases parasitic drag and reduces $C_{L,max}$
- 2.7 degree washout optimal for this wing

Taper Ratio	Sweep (deg) 26	27	28	29	30	31
0.27	396.98	396.80	396.64	396.61	396.61	396.64
0.28	396.76	396.55	396.45	396.42	396.42	396.45
0.29	396.58	396.40	396.30	396.27	396.28	396.31
0.3	396.50	396.32	396.23	396.17	396.20	396.22
0.31	396.71	396.53	396.43	396.39	396.39	396.41
0.32	396.68	396.49	396.39	396.35	396.35	396.37

Wing Drag Coefficient (in drag counts) vs. Taper Ratio and Sweep Angle

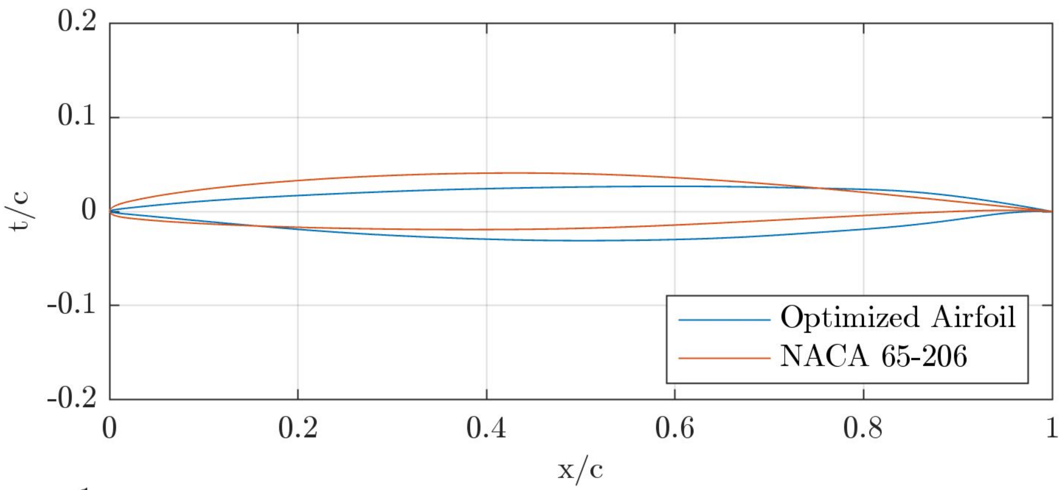
Airfoil Design

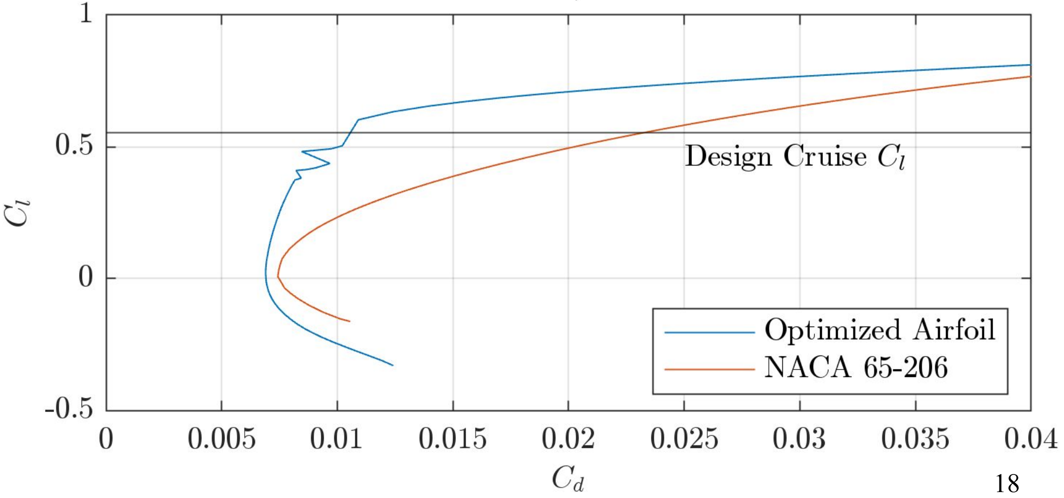
Multi-Point Design Optimization

- Drag reduction at cruise and supersonic dash operating conditions

Results

- Flattened top surface and aft camber reduces cruise wave drag
- Thin leading edge reduces supersonic drag
- Sectional drag reduced by 127 counts





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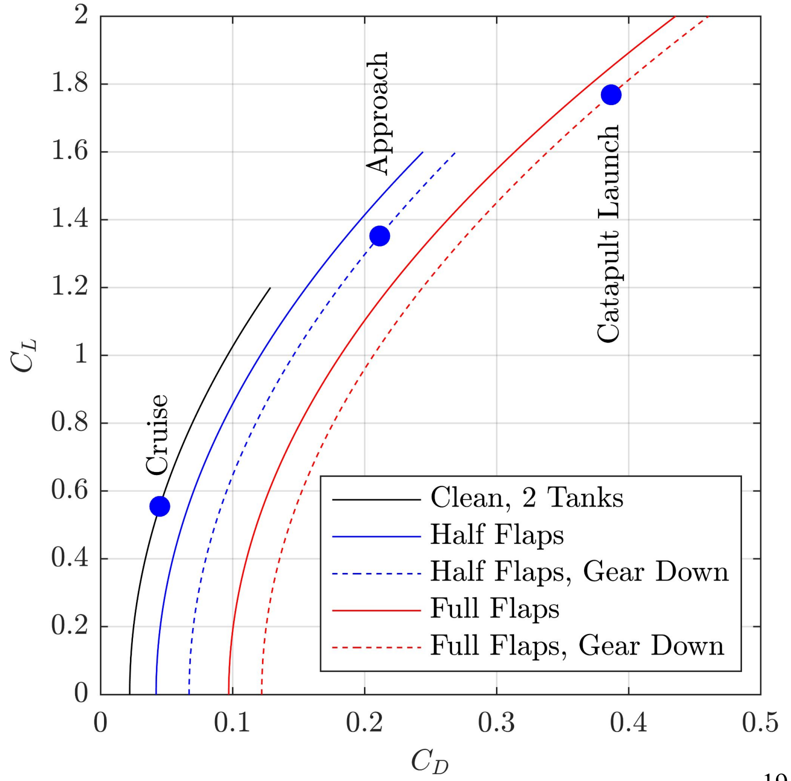
Drag Polars

Assumptions

- Parameters determined using historical values
- Subsonic C_{D0} listed
- Used for preliminary sizing

Polar	C_{D0}	e	$C_{L,max}$
Clean, 2 external tanks	0.0221	0.80	1.2
Half flaps	0.0421	0.75	1.6
Half flaps, gear down	0.0671	0.75	1.6
Full flaps	0.0971	0.70	2.0
Full flaps, gear down	0.1221	0.70	2.0



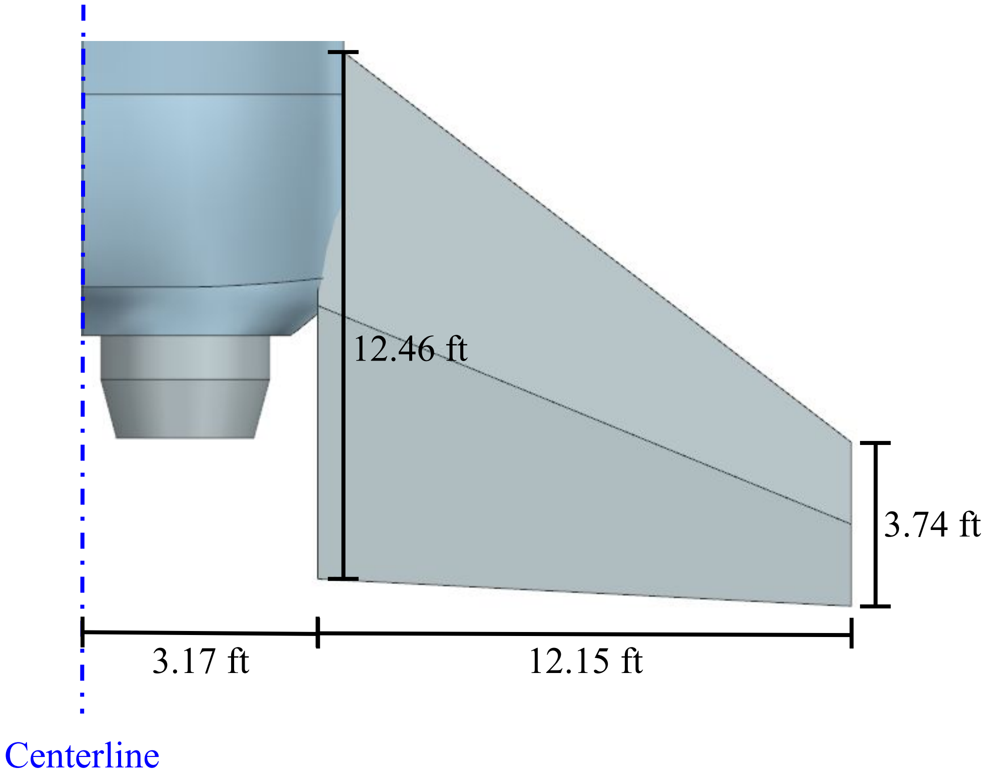
Empennage Design

Sizing

- Raymer tail volume coefficients
  - $c_{HT} = 0.36$
  - $c_{VT} = 0.07$
- Swept to achieve 5 degree higher LE sweep than main wing

Dimensions

- Horizontal Tail
  - Moment Arm Length: 15 ft
  - Taper Ratio: 0.3
  - Aspect Ratio: 3
- Vertical Tail
  - Moment Arm Length: 12.5 ft
  - Taper Ratio: 0.5
  - Aspect Ratio: 1



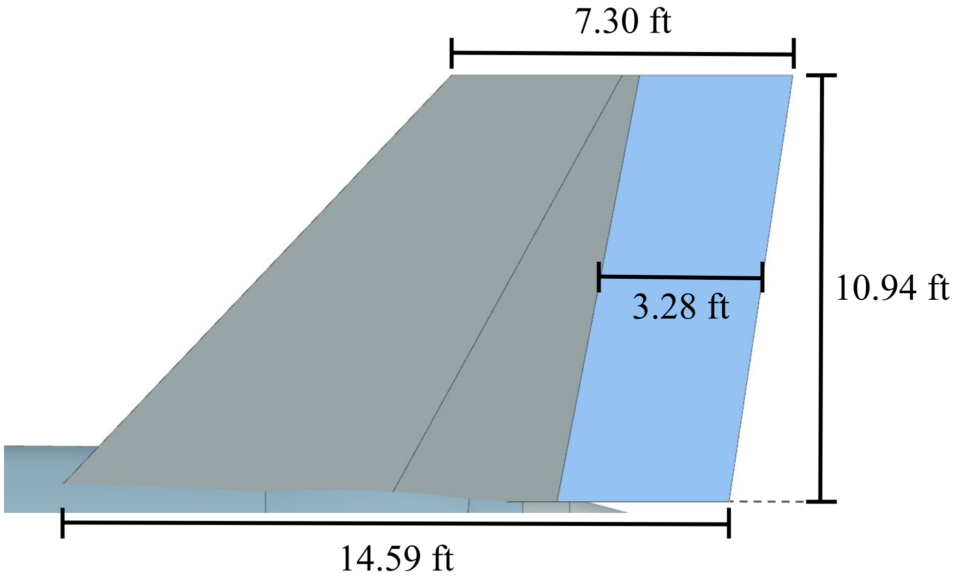
Control Surfaces: Elevator and Rudder

Elevator

- All-moving stabilator
  - Improves maneuverability
  - Reduces c_{HT} by 10%

Rudder

- Constant width
  - Improves manufacturability
- 30% average chord fraction
  - 22.5% inboard chord fraction
  - 45% outboard chord fraction



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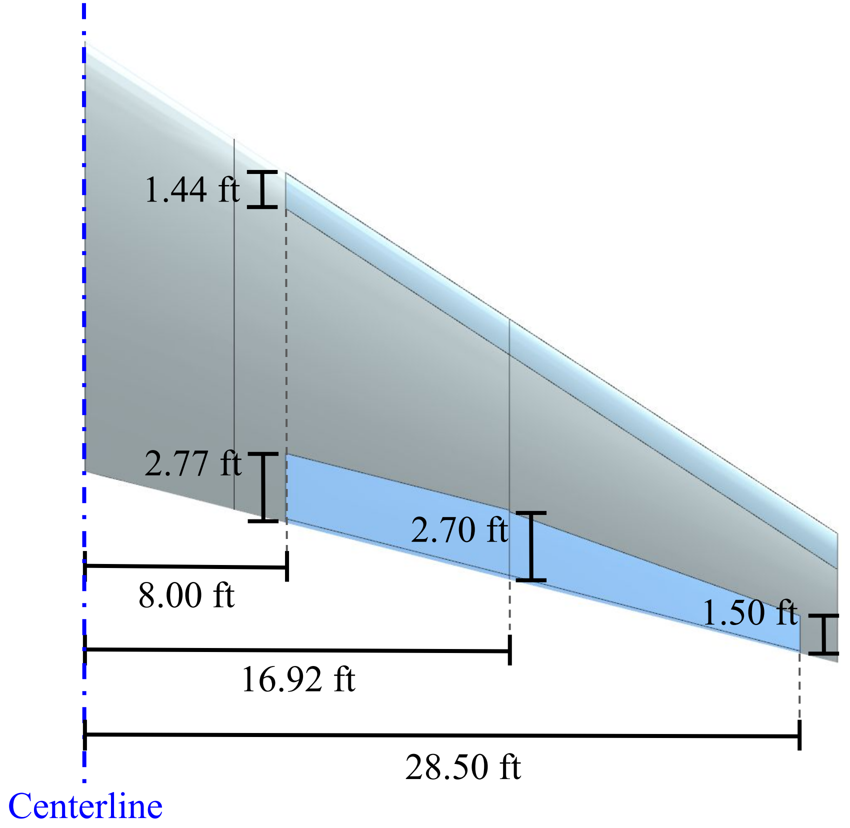
Control Surfaces: Flaps and Ailerons

Flaps

- Leading edge flaps
  - Span: 26% to 100%
  - Constant width, 15% average chord fraction
- Trailing edge slotted flaps
  - Span: 26.66% to 56.39%
  - Constant width, 25% chord fraction
  - Increase takeoff $C_{L,max}$ for catapult launch

Ailerons

- Span: 56.39% to 95%
  - Begins at hinge point
- Constant chord hinge, 26% average chord fraction
  - Avoids large chord fraction at tip
  - Matches Raymer recommendations



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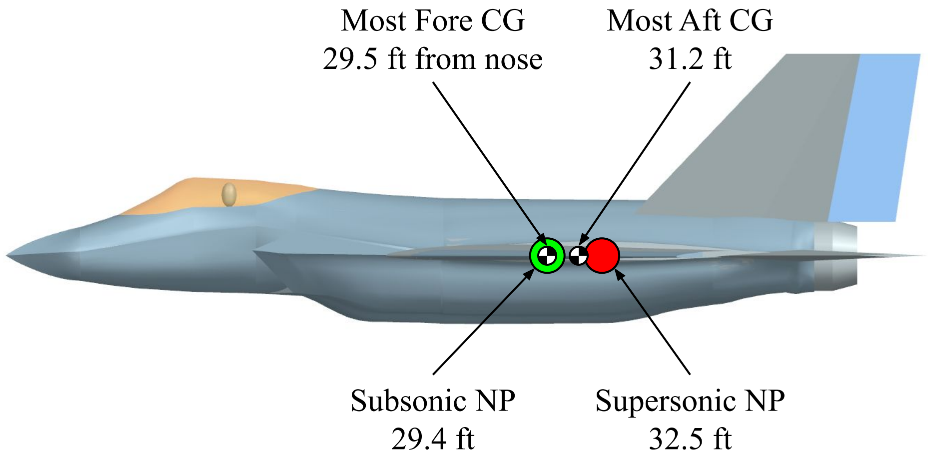
Static Margin and Stability

Subsonic	Air to Air Static Margin (%MAC)	Strike Static Margin (%MAC)
Empty	-12.3	-12.3
Payload	-9.6	-7.6
Payload and Rear Fuselage fuel	-15.0	-13.1
Payload and Internal Fuel	-5.3	-4.1
Payload, Internal, and External Fuel	-1.8	-0.8



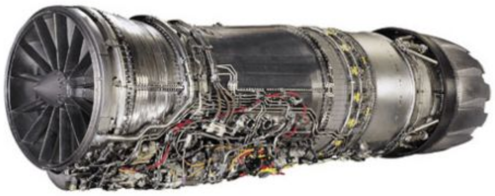
Add 25.2% for supersonic SM

NO THAI GRUBBIN

Power Plant

2x F110-GE-132 Engines

- Maximum Thrust: 33,093 SLS-lb per engine
- Military Thrust: 17,669 SLS-lb per engine
- Existing production design: used on F-16E/F
- Meets design thrust requirements with a lower dry TSFC (0.68 lb/hr/lbf) than other comparable engines



F110-GE-132 Engine

Thrust Corrections

- Accounts for 10% installed thrust reduction due to offtakes
  - Inlet Pressure Loss - 1.35%
  - Engine Bleed - 8%
- Accounts for atmospheric thrust lapse and ram effects

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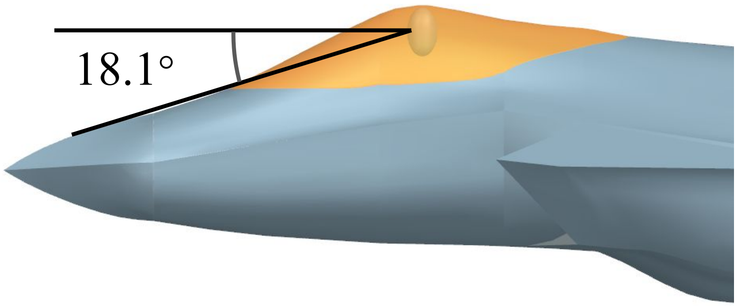
Cockpit Layout

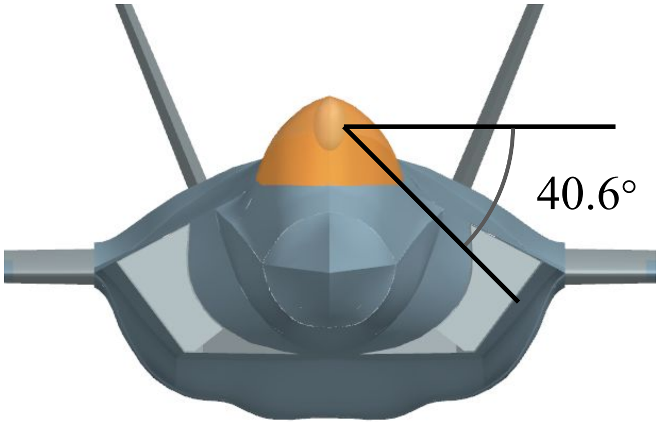
Crew

- 1 crew member
  - Reduced weight
  - Reduced volume

Pilot View Angles

- Overnose angle: 18.1°
- Over-the-side angle: 40.6°





Weapons Bay

Internal Storage

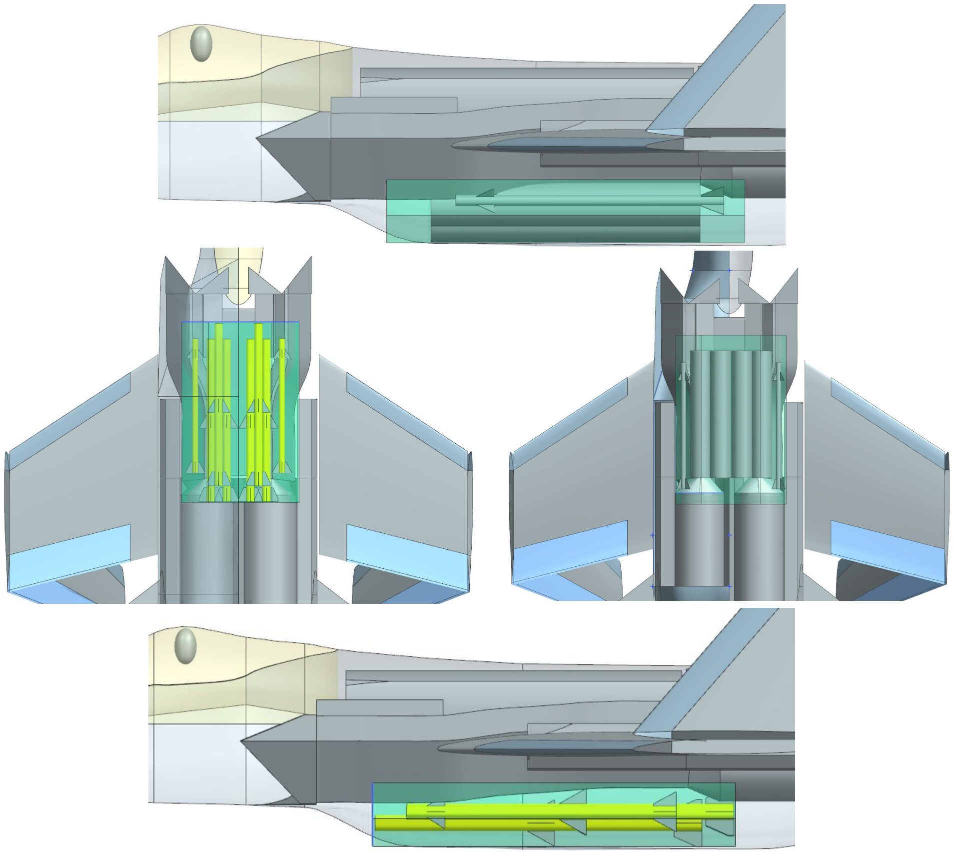
- Payload fully stored internally for reduced radar cross section and less drag
- Located at 36% to 63% along the fuselage length

Air to Air Mission Payload

- 6x AIM-120C
- 2x AIM-9X

Strike Mission Payload

- 4x MK-83 JDAM
- 2x AIM-9X



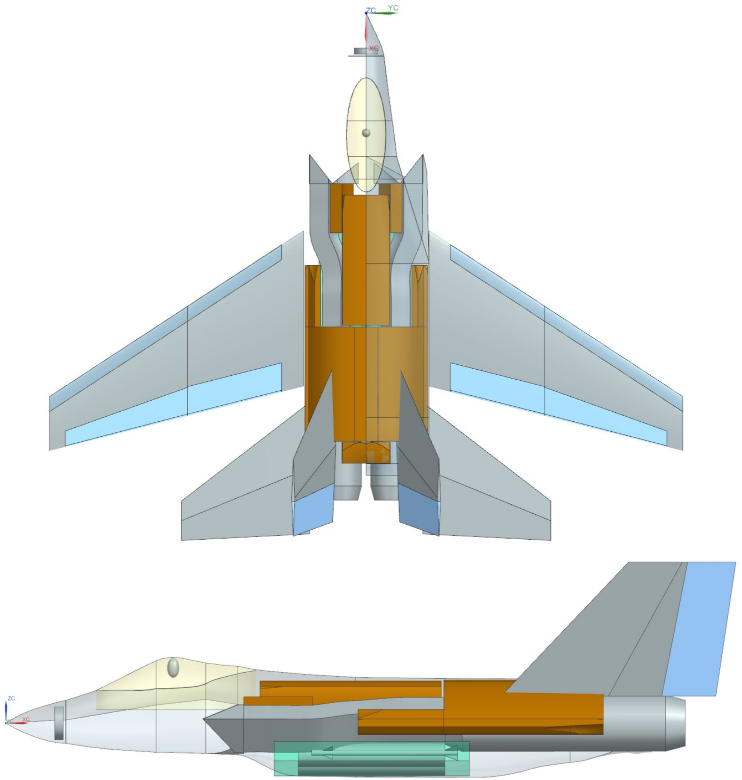
Fuselage Weapons Bay with Air to Air Payload in Yellow and Strike in Gray

[2-5]

NO THAI GRUBBIN

Fuel Tanks

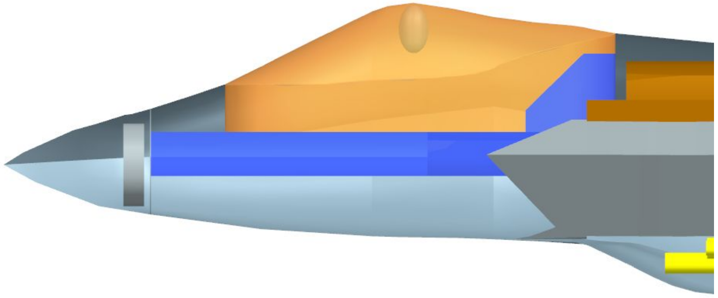
Location	Fuel Volume (ft^3)	Fuel Weight (lbs)
2x 480 gallon external tanks	128.3	6,470
Wings	92.8	4,678
Vertical Tails	24.6	1,242
Fuselage	394.4	19,884
Total	640.1	32,274



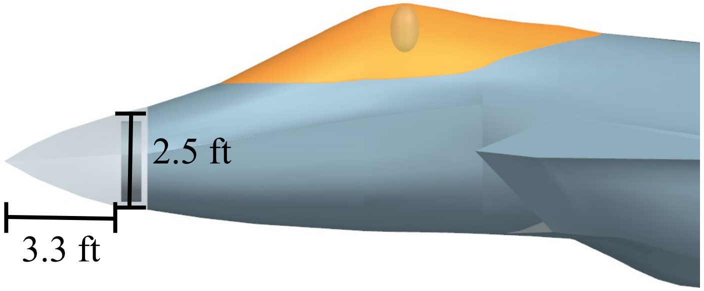
Fuselage Fuel Tanks (in brown) located at 32% to 85% of the fuselage length

Avionics and Subsystems

- Avionics Bays Volume: 61.1 ft^3
- Avionics placed near the cockpit for wire length
- AN/APG-81 Radar
  - Enemy tracking and ordnance guidance
- JPALS Approach & Landing System
  - Precision guidance for landing on carriers
- Quadruple redundant fly-by-wire system
  - Ensures aircraft is stable while being maneuverable
- AN/ALQ-214 IDECM
  - Counters enemy missiles with jammers and decoys
- Advanced Mission Computer
  - Modular system that processes and controls a lot of avionics



Avionics Bays (in blue)



AN/APG-81 Radar Location

[6-9]

Cost Breakdown (2025 USD)

Unit Flyaway Cost for 500 aircraft: 105 $M
Flyaway Cost for 500 aircraft: 52.5 $B

Engine Price ($M)	Avionics Price ($M)	A2A Ordnance Price ($M)	Strike Ordnance Price ($M)	DOC ($M)	DOC/pay-nmi ($)
18.8	28.0	6.63	1.33	7.38	0.0146

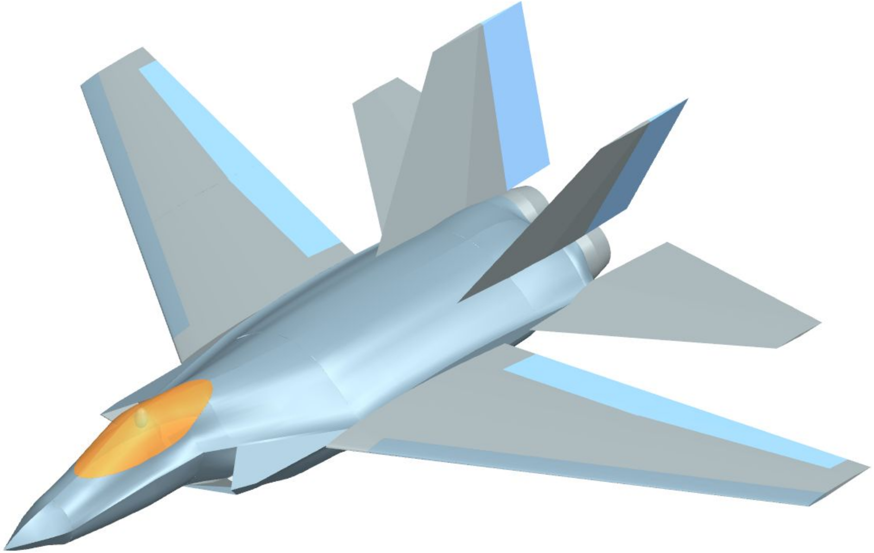
Notes

- F/A-18E/F Unit Flyaway Cost for 552 aircraft: 113 $M
- F/A-18E/F Flyaway Cost for 500 aircraft: 56.5 $B
- While this meets the RFP requirements of breaking the cost trend, further investigation into empty weight and avionics needs to be done into why the resulting F/A-67 unit flyaway cost is cheaper by 8 $M

Next Steps

Refined Analysis

- Weights
  - Component buildup
  - Empty CG prediction
- Aerodynamics
  - Drag buildup
  - Trimmed Analysis
  - Stability
- Structures
  - Landing gear sizing
  - Flight loads and envelope



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References

[1] Andrew B. Lambe and Joaquim R. R. A. Martins. “Extensions to the Design Structure Matrix for the Description of Multidisciplinary Design, Analysis, and Optimization Processes”. In: Structural and Multidisciplinary Optimization 46 (2012), pp. 273–284. doi: 10.1007/s00158-012-0763-y.

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NO THAI GRUBBIN

Backup Slides

List of Symbols

T/W	Thrust-To-Weight Ratio
W/S	Wing Loading (lbs/ft^2)
T	Thrust (lbs)
S	Wing reference area (ft^2)
AR	Aspect Ratio
L/D	Lift-To-Drag Ratio
$C_{L,max}$	Maximum lift coefficient
MAC	Mean Aerodynamic Chord (ft)
$C_{D,ind}$	Induced drag coefficient
$C_{D,visc}$	Viscous drag coefficient
$C_{D,wave}$	Wave drag coefficient
C_{HT}	Horizontal tail volume coefficient
C_{VT}	Vertical tail volume coefficient
$C_{N\beta}$	Yawing moment derivative with respect to sideslip
CG	Center of gravity
SLS	Sea-level-static
$CD0$	Zero-lift drag coefficient

NO THAI GRUBBIN

Detailed Cost Breakdown

	Cost (2025 USD)		Auxiliary data (2025 USD)		Environmental, Fleet Compatibility, & Weapons (2025 USD)
Crew	$1.11 Million	**Block time [hr]**	Strike: 4.73 Air-Air: 4.11	**JDAM Price [USD] MK-83 JDAM (each)**	$40,017
Avg Weapons Expenditure	Strike: $6.63 Million Air-Air: $1.33 Million	**Fuel density [lb/gal] JP-5**	6.74	**Missile Price [USD] AIM-120C (each)**	$910,960
Fuel	$1.38 Million	**Fuel price [USD/gal] JP-5**	$4.05	**Missile Price [USD] AIM-9X (each)**	$583,243
Oil	$6,905	**Lubricating oil density [lb/gal]**	7.2	**Life Support, etc. [USD]**	Ejection Seat: $376,350 Helmet: $412,002
Airport Fees	Not Considered	**Lubricating oil price [USD/gal]**	$14.58	**Avionics [USD]**	$28 million

Detailed Cost Breakdown

	Cost (2025 USD)		Auxiliary data (2025 USD)		Environmental, Fleet Compatibility, & Weapons (2025 USD)
Navigation Fees	Not Considered	**Labor rate [USD/hr]**	Engineering and Pilots: $161 Manufacturing: $137 Maintenance: $118	**Surface Treatments [USD/year]**	$203,125
Airframe Maintenance	$2.19 Million	**Aircraft price [USD]**	$105.5 million	**Subsystem Technologies [USD]**	Not Considered
Engine Maintenance	$778,000	**Total engine price [USD]**	$18.8 million		
		Payload [lbm]	Strike: 4,424 Air-Air: 2,460		
COC	$7.48 Million	**Range [nmi]**	Strike: 2400 Air-Air: 1800		

NO THAI GRUBBIN

Detailed Cost Breakdown

	Cost (2025 USD)		Auxiliary data (2025 USD)		Environmental, Fleet Compatibility, & Weapons (2025 USD)
COC/pay-nmi	$0.01510	**Missiles per mission**	Strike: 2 Air-Air: 8		

Full Aircraft Comparison Tables

Aircraft	Takeoff Weight (lbs)	Empty Weight (lbs)	Air to air mission payload (lbs)	Strike mission payload (lbs)	Combat Radius (nmi)
F/A-67	76366	41432	2460	4424	900-1020
F/A-18E	66000	32081	2946	5442	444-462
F-35C	70000	34800	up to 18000	up to 18000	670
F-15E	81000	37500	2460	4398	687

Aircraft	Air to air mission fuel burn (lbs)	Strike mission fuel burn (lbs)	Aircraft Length (ft)	Cruise L/D	T/W	W/S (lbs/ft^2)
F/A-67	29411	27480	50	12.4	0.780	114
F/A-18E	18218	18313	60.1	-	0.651	132
F-35C	18600	18600	51.5	-	0.614	105
F-15E	22300	22300	63.8	-	0.728	133

Full Aircraft Comparison Tables

Aircraft	Engine Type	Maximum sea-level static thrust (per engine)	Engine Cruise SFC
F/A-67	F110-GE-132	33093	0.68
F/A-18E	F414-GE-400	21496	0.82
F-35C	F135-PW-400	43000	-
F-15E	F110-GE-129	29474	0.67

Aircraft	Span (ft)	Reference Area (ft^2)	Aspect Ratio	Average wing t/c	Cruise Mach number (maximum)	Cruise Mach number (economic)
F/A-67	60	670	5.4	5.70%	2.3	0.84
F/A-18E	44.7	500	3.5	-	1.6	0.84
F-35C	43	668	2.7	-	1.6	-
F-15E	42.8	608	3.0	-	2.5	-

NO THAI GRUBBIN

Aircraft Comparison Tables

Aircraft	Static margin (Subsonic, Supersonic)	Maximum landing distance (ft)	Maximum takeoff distance (ft)	Flyaway cost for unit production run of 500 (USD 2025)
F/A-67	-15%, 10.2%	6000	2500	52.5 billion
F/A-18E	-	8500	4,666	56.5 billion
F-35C	-	-	-	63.5 billion
F-15E	-	12,000	11,000	30.7 billion

Notes

- For landing, assume takeoff weight, altitude of 4,000 ft, temperature of 89F

Installed Thrust from Pressure Loss and Bleed

$$\%_{TL_{PR}} = c_r[\eta_{\text{ref}} - \eta_{\text{actual}}] * 100\%$$

$$\%_{TL_{EB}} = C_b \frac{\dot{m}_b}{\dot{m}_e}$$

$$T_{\text{installed}} = T(1 - \frac{\%_{TL_{PR}} + \%_{TL_{EB}}}{100})$$

Thrust Lapse

$$\theta = T/T_{\text{std}} \quad \text{and} \quad \theta_0 = \theta \left(1 + \frac{\gamma - 1}{2} M^2\right)$$

$$\delta = P/P_{\text{std}} \quad \text{and} \quad \delta_0 = \delta \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\frac{\gamma}{\gamma - 1}}$$

$$\text{Maximum:} \begin{cases} \alpha = \delta_0, & \theta_0 \leq TR \\ \alpha = \delta_0(1 - 3.5(\theta_0 - TR)/\theta_0), & \theta_0 > TR \end{cases}$$

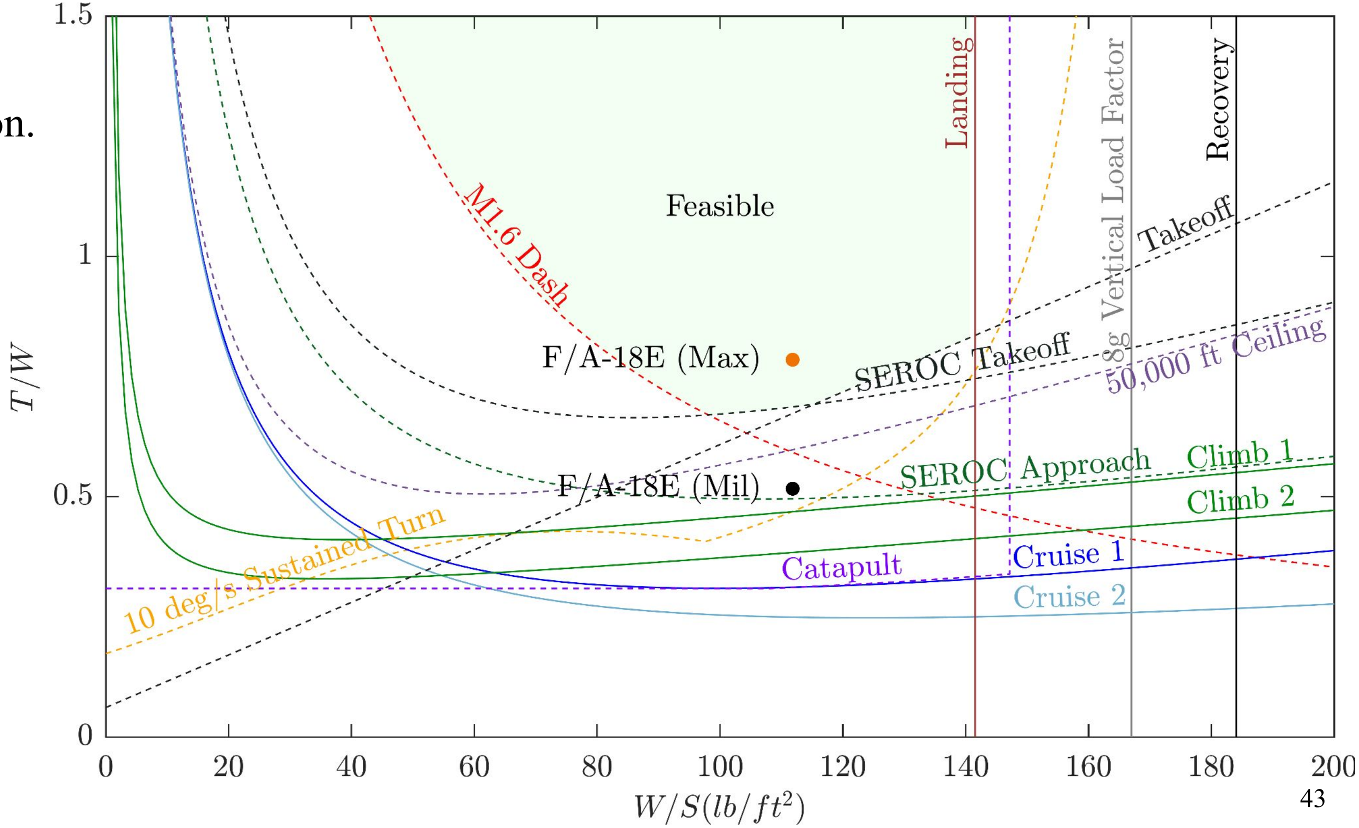
$$\text{Military:} \begin{cases} \alpha = \delta_0, & \theta_0 \leq TR \\ \alpha = \delta_0(1 - 3.8(\theta_0 - TR)/\theta_0), & \theta_0 > TR \end{cases}$$

Aircraft Component Locations Table

Component	% of Fuselage Length
Start of Engines	62%
Cockpit	12.3-33.9%
Pilot	22.7%
Avionics	8.2-33.9%
Radar	6.6%
Fuel Tanks	32-85%
Weapons Bay	36-63%
External Fuel Tanks	50%

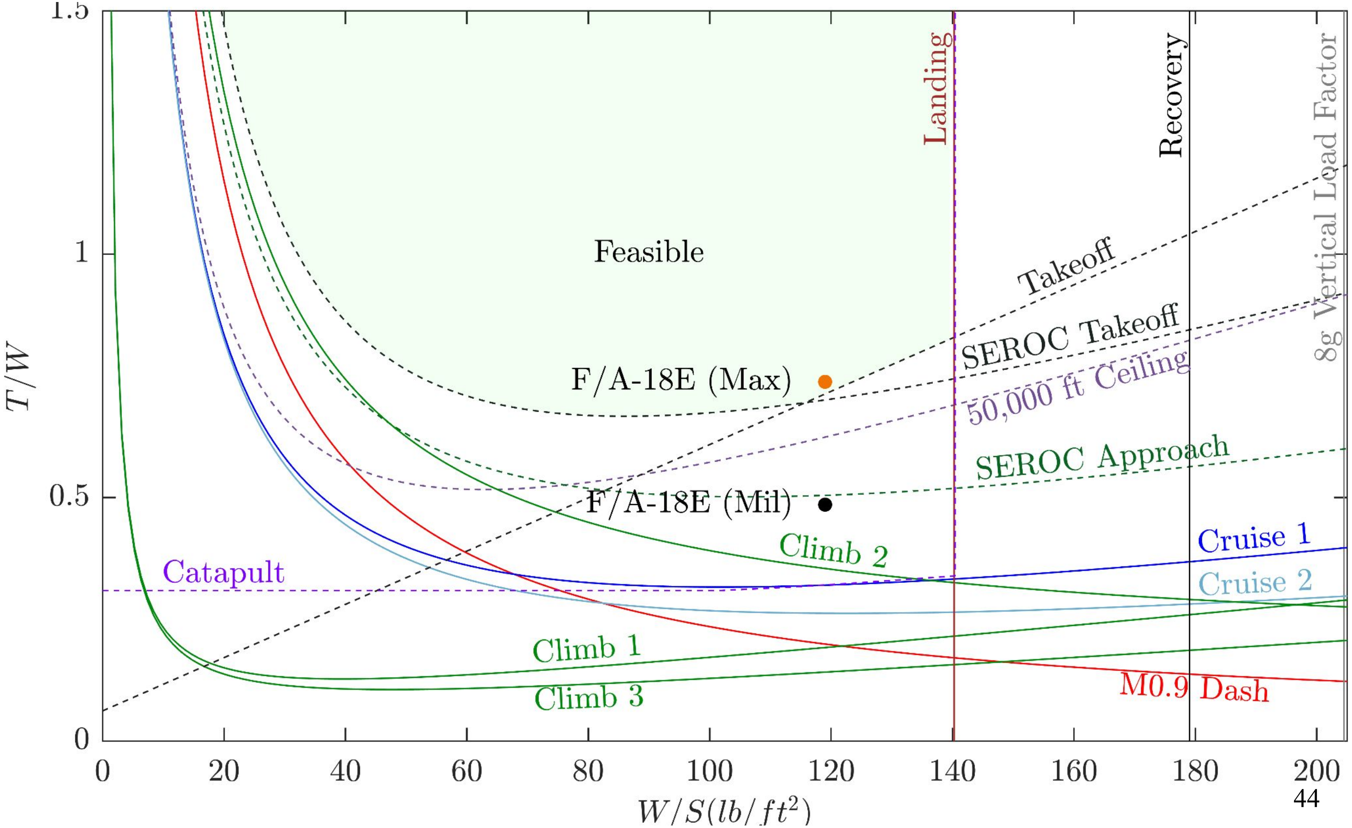
Sizing plot validation (air to air)

F/A-18E design point is feasible for air to air mission.



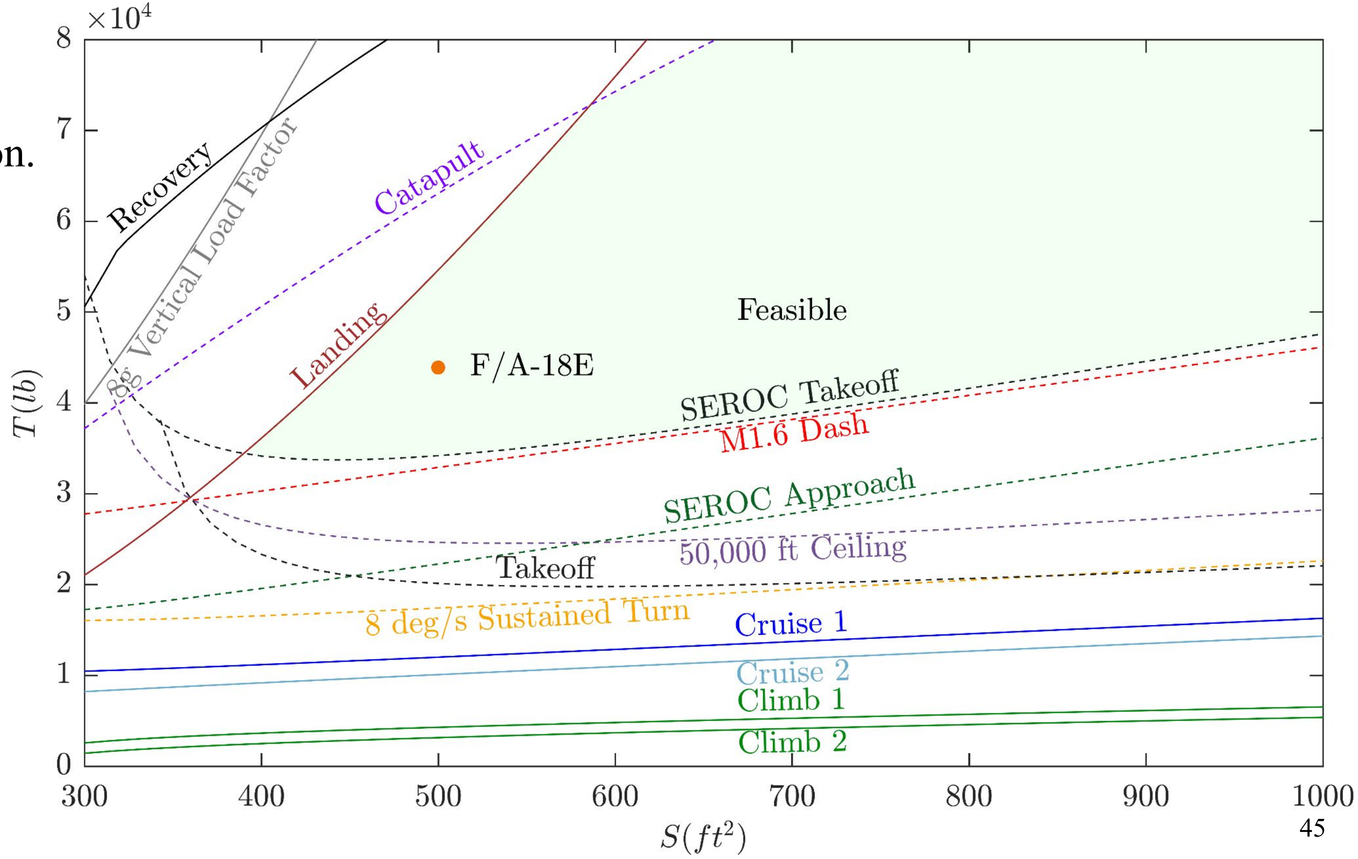
Sizing plot validation (strike)

F/A-18E design point is feasible for strike mission.



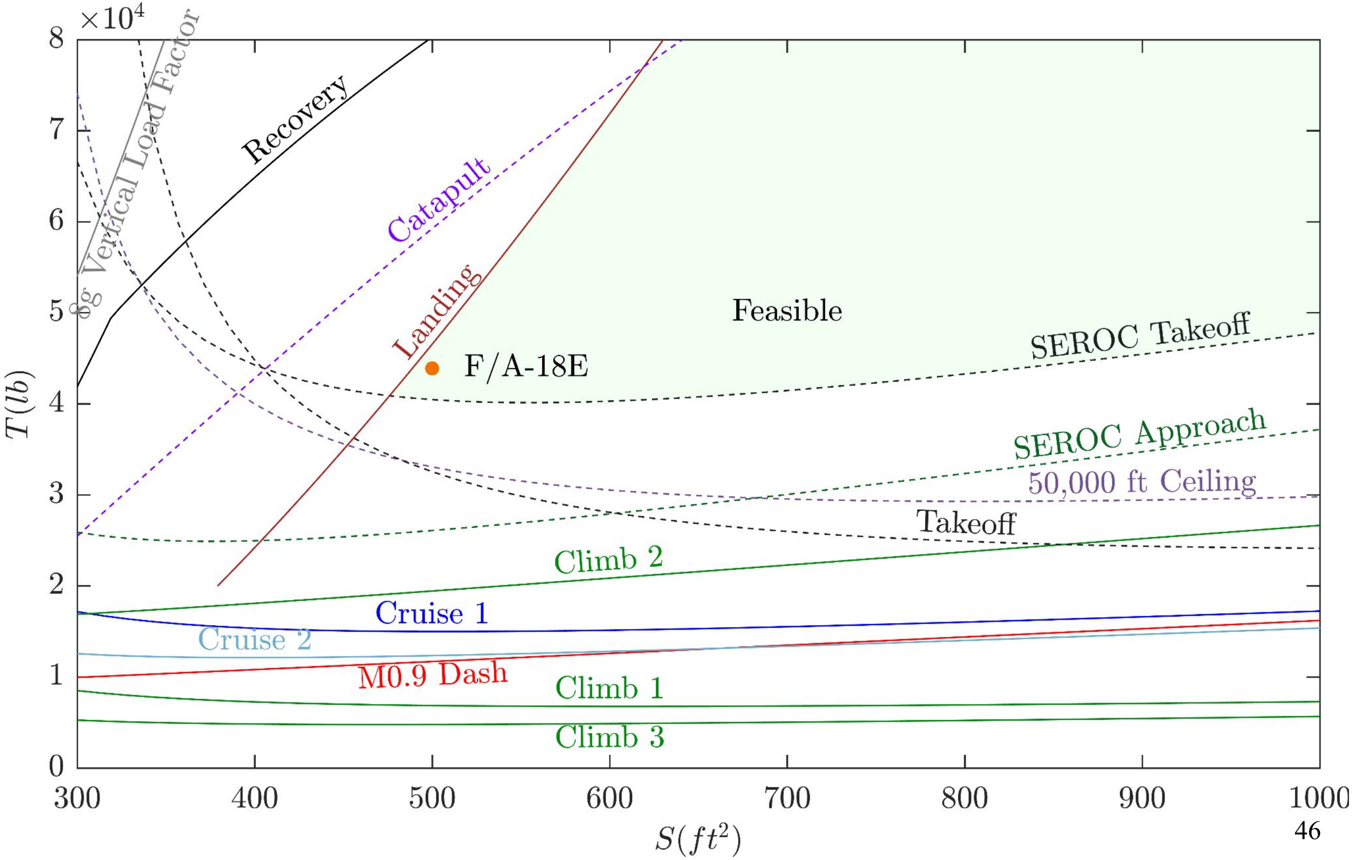
T-S plot validation (air to air)

F/A-18E design point is feasible for air to air mission.



T-S plot validation (strike)

F/A-18E design point is feasible for strike mission.



Initial Weight Estimate from Historical Regressions

	Air to air	Strike
Empty weight (lbs)	36,570	23,830
Payload weight (lbs)	2,460	2,398
Fuel weight (lbs)	40,800	24,420
Takeoff weight (lbs)	80,040	50,840

NO THAI GRUBBIN

Initial Weight Estimate from Initial Weight Code

	Air to air	Strike
Empty weight (lbs)	43,379	43,341
Payload weight (lbs)	2,460	4,424
Fuel weight (lbs)	34,465	32,458
Takeoff weight (lbs)	80,504	80,423

Weight Estimate from T-S Plot Design Point

	Air to air	Strike
Empty weight (lbs)	43,117	43,254
Payload weight (lbs)	2,460	4,424
Fuel weight (lbs)	34,169	32,360
Takeoff weight (lbs)	79,946	80,237

Planform Trade Study Results

e		Sweep (deg)					
		26	27	28	29	30	31
Taper Ratio	0.27	0.98888	0.98876	0.98845	0.98831	0.98816	0.98801
	0.28	0.98898	0.98903	0.98892	0.98879	0.98865	0.9885
	0.29	0.9889	0.98878	0.98864	0.9885	0.98834	0.98818
	0.3	0.98879	0.98866	0.98851	0.98907	0.98932	0.98919
	0.31	0.98978	0.98969	0.98959	0.98947	0.98934	0.98921
	0.32	0.98995	0.98988	0.98979	0.98969	0.98959	0.98946

Oswald’s efficiency

Planform Trade Study Results Cont'd

CDp		Sweep (deg)					
		26	27	28	29	30	31
Taper Ratio	0.27	120.5	120.5	120.4	120.4	120.4	120.4
	0.28	120.3	120.3	120.3	120.3	120.3	120.3
	0.29	120.1	120.1	120.1	120.1	120.1	120.1
	0.3	120	120	120	120.1	120.2	120.2
	0.31	120.4	120.4	120.4	120.4	120.4	120.4
	0.32	120.4	120.4	120.4	120.4	120.4	120.4

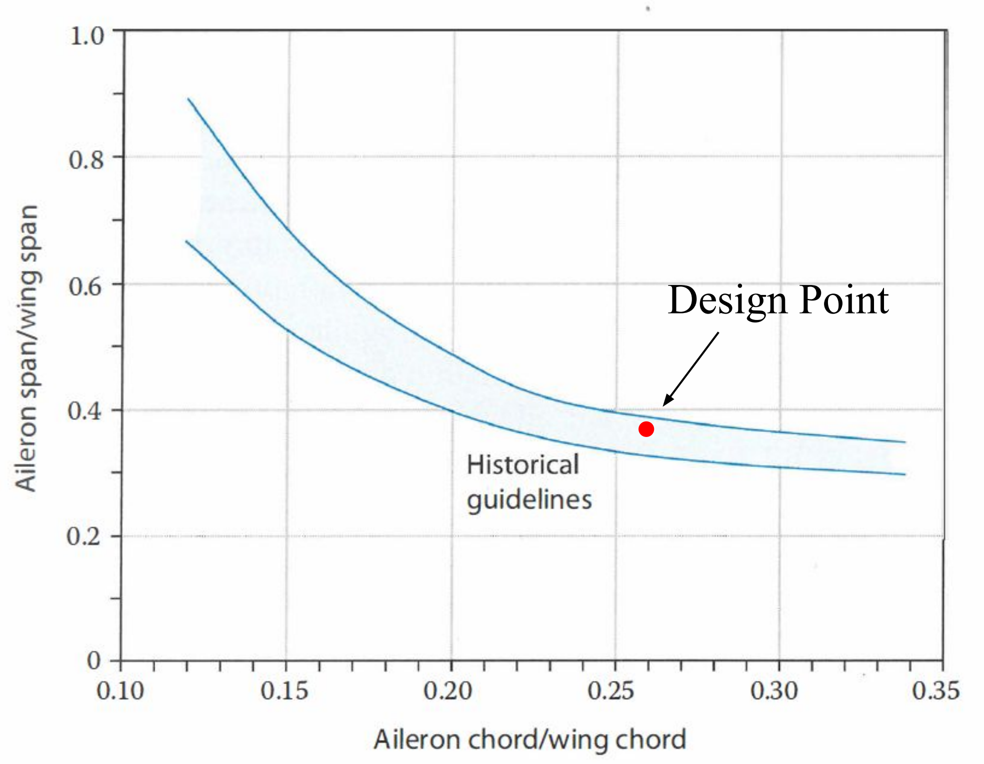
Parasitic drag coefficient (drag counts)

CDwave		Sweep (deg)					
		26	27	28	29	30	31
Taper Ratio	0.27	0.40429	0.202672	0.084297	0.025834	0.00437	0.000133
	0.28	0.40429	0.202672	0.084297	0.025834	0.00437	0.000133
	0.29	0.40429	0.202672	0.084297	0.025834	0.00437	0.000133
	0.3	0.40429	0.202672	0.084297	0.025834	0.00437	0.000133
	0.31	0.40429	0.202672	0.084297	0.025834	0.00437	0.000133
	0.32	0.40429	0.202672	0.084297	0.025834	0.00437	0.000133

Wave drag coefficient (drag counts)

NO THAI GRUBBIN

Aileron Sizing



Raymer Fig. 6.3